

CareerBuddy: AI-Driven Career Transformation

Technical Implementation Notes

This document provides context regarding the technical architecture and development methodology used for the *CareerBuddy* prototype during the Dean's Open Innovation Challenge 2026.

Development Methodology: AI-Assisted Rapid Prototyping

To meet the rigorous deadlines of the innovation challenge while delivering a high-fidelity "Digital Twin" experience, I utilized an **AI-augmented development workflow** via **Atoms.dev**.

- **Architectural Role:** My role focused on defining the system requirements, the user-persona logic, and the data-mapping between market signals and the "Digital Twin" interface.
- **Implementation:** The code was generated using high-level design prompts and logic definitions, then refined to ensure the "Skill-based Graph" and "Forecast Engine" components were visually and functionally accurate.

Folder Structure & Key Files

As seen in the repository (and original development environment image_0162b8.png):

- **src/:** Contains the core React components. This is where the logic for the "Digital Twin" persona and the career roadmap visualization resides.
- **tailwind.config.ts:** Defines the custom design system. I configured this to ensure the "Career Buddy" branding was consistent across all responsive views.
- **vite.config.ts:** The build tool configuration used to ensure fast hot-reloading and optimized deployment.
- **components.json:** A blueprint of the modular UI elements used to build the dashboard.

Key Engineering Challenges Addressed

1. **Data Visualization:** Translating complex predictive market data into a "GPS Learning Roadmap" that a non-technical user can understand.
2. **Persona Mapping:** Structuring the interface to handle three distinct user flows: Students, Professionals, and Specialists.
3. **Proactive Interaction:** Moving away from standard search bars to a "Digital Twin" partner model, which required a specific UI layout for AI-to-User collaboration.

Future Technical Roadmap

If this project were moved to a full production environment, the next steps would involve:

- Connecting the frontend to a live **PostgreSQL** database for persistent "Digital Twin" storage.
- Integrating real-time **LLM APIs** to power the proactive career advice chat.

- Expanding the **Forecast Engine** with more granular patent and scientific data scraping.